# allocsys

Backend Developer — APIs, Infra & AI-Agent Tooling

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## SUMMARY

Backend-focused developer who builds the infrastructure layer AI agents actually run on — auth, rate limiting, multi-service integrations — rather than just wiring prompts together. Comfortable owning a project end-to-end: designing the API surface, hardening it (timing-safe auth, IP allowlisting, proxy-aware rate limiting), and documenting the reasoning so someone else can maintain it. Recent focus: Model Context Protocol (MCP) servers and agentic workflow automation (n8n + LangChain), plus root-causing bugs in real-world open-source infrastructure across JavaScript, Python, and Rust codebases. Ships full-stack TypeScript apps on Cloudflare Workers.

## PROJECTS

### madmcp

*Node.js / Express · MCP server · Redis*

- Built and deployed an MCP server exposing ~50 tools across 8 connectors (GitHub, Cloudflare D1/KV/R2/Workers, Notion, Mem0, generic fetch, Exa, Gemini) to any MCP-compatible AI client.
- Designed a delegation-first architecture: one tool hands an open-ended, multi-step investigation to a second model that runs its own tool loop server-side, with Redis-backed checkpointing so a rate-limited run resumes from its last step instead of restarting.
- Separated tools that touch untrusted external content (web research, fetch) from those that touch internal systems (GitHub, Cloudflare, Notion), so external data can't reach credentials.
- Designed the auth layer: timing-safe key comparison, IP allowlisting scoped to Claude's connector range, and per-route rate limiting, ordered so a leaked key alone can't get in. Handled proxy-hop IP trust, large-push body limits, and per-connector token failure; PR-based workflow with GitHub Actions CI, vitest and supertest.

### proetsy

*Node.js · React · SQLite · Electron · Playwright*

- Built a local-first pipeline that turns artwork into ready-to-paste Etsy listings and mockups: optional image analysis, listing generation, PSD-template mockups, and a human-in-the-loop review step (publishing stays manual by design).
- Built an LLM provider layer with Gemini primary and Claude fallback, key store, queue and rate limits, plus an embedding-based taste filter in pure JS/WASM (runs on Termux).
- Tested every module (unit, idempotency, route-level integration) and added Playwright E2E against a deterministic fixture LLM; shipped a CI-verified Electron Windows installer.

### news-market-ai

*Cloudflare Workers · D1 · KV · Gemini · Next.js*

- Built a news-to-trade-signal pipeline: ingestion from free sources and custom scrapers, multi-agent LLM analysis, and signal generation, designed around free-tier Worker limits.
- Wrote a point-in-time backtester and documented rollout gates (backtest → paper → micro-live); live trading is deliberately off.
- Built a multi-key Gemini model-cascade client with KV cooldown state, and a mobile-friendly Next.js 16 ops dashboard served as a Worker with login and a session-gated API.

### ai-campaign-builder

*TypeScript · React · Cloudflare Workers · D1*

- Built a monorepo where a business owner states a goal and an AI agent generates a loyalty or gamification campaign. Apps: owner dashboard, customer app, staff POS, review console, public microsites, and a backend API.
- Shared api-client, UI-kit and campaign-editor packages; JWT auth, D1 migrations, Vitest tests, and CI deploys that provision D1 and KV automatically.

### peakplast

*Cloudflare Workers · Telegram · Gemini · Zarinpal*

- Built and run a production Telegram ordering bot: Gemini voice-order extraction with multi-key fallback, Zarinpal payment links and verification, an admin panel, and a REST API.
- Handled concurrency and abuse (per-chat locking, voice flood limits, address de-duplication). Planning and executing a safe split of the ~7,000-line single-file Worker into ES modules, one verified, deployable step at a time.

### clothes-shop

*Next.js · TypeScript · Tailwind CSS · RTL*

- Built the front-end of a Persian (right-to-left) online clothing shop: home, shop, product, cart, account and content pages, with a mobile tab bar and configurable site, theme and product data. Backend, checkout and payments are planned next.

### Agentic Email Triage & Ops Automation

*n8n · LangChain · RAG*

- Designed an n8n workflow where an AI agent classifies inbound email across Gmail, Telegram and WhatsApp, answers from a RAG knowledge base, and drafts replies only for non-urgent cases.
- Put the safety boundary in the system prompt: escalation-worthy emails are flagged for a human and never auto-replied to. Used an OpenAI → Gemini → Claude fallback chain and forced JSON output so Sheets, CRM and LINE nodes route deterministically; tested on 6 scenarios.

### Developer tools (Termux / Android)

*Bash · Node.js · Playwright · Android*

- **cf-manager:** menu-driven Cloudflare admin panel in one bash script: Workers deploy/rollback, KV/D1/R2/Hyperdrive, OAuth2 + PKCE with multi-account switching, GitHub sync.
- **logger (DOM mapper):** maps a live DOM, network traffic and replayable actions via remote Browserless.io, working around Playwright's lack of Android support.
- **ezgit** (Git helper that redacts secrets before diffs reach an LLM) and **openhands-android-client** (native client for OpenHands coding agents).

## OPEN SOURCE CONTRIBUTIONS

### transact-rs/sqlx

*Rust · PR #4351 · open*

- Fixed the *Any* driver emitting *?* placeholders on Postgres (issue #3000) by adding a SQL-aware rewrite to *$N* that skips string literals, quoted identifiers, dollar-quoted strings, and comments, applied on the Postgres path only so MySQL and SQLite are unaffected. CI passes on the fork.

### cloudflare/workers-sdk

*4 PRs · 1 merged, 2 closed, 1 open*

- Root-caused a Windows-only crash from non-ASCII paths (Latin-1 header violation) across 5 call sites — merged.
- Fixed a false infinite-redirect-loop deploy warning — closed, not merged.
- Fixed local Images binding transform parity (fit/gravity/background) — original PR closed after an accidental fork deletion mid-review; restored and continued in a follow-up PR, currently open and in review.

### urllib3/urllib3

*PR #5121 · closed (superseded)*

- Replaced a deprecated pyOpenSSL X.509 API call with the cryptography library equivalent, with new tests for both a present and missing commonName.

## SKILLS

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|---|---|
| **Languages** | JavaScript / TypeScript, Node.js, Python, SQL, Bash; Rust (open-source contributions) |
| **Backend & APIs** | Express, REST API design, OAuth2 / PKCE, JWT, rate limiting, IP allowlisting, webhooks, React, Next.js, Tailwind, Electron |
| **AI / Agent Tooling** | Model Context Protocol (MCP), LangChain, RAG / vector DBs (Qdrant), n8n, multi-provider LLM fallback, structured output, prompt design |
| **Infrastructure** | Git, Cloudflare (D1, KV, R2, Workers, Hyperdrive), Redis, SQLite, Render / Vercel, GitHub Actions CI, Vitest, Playwright, Termux / Android |